

# The Homology Cliff in Frozen ESM-2 Biosecurity Retrieval: Characterization, Pre-Registered Null Controls, Cross-Family Mechanism, and a Deployable Rescue

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## Abstract

Frozen protein language models (PLMs) increasingly feed biosecurity retrieval pipelines. We characterize a systematic failure mode, the *homology cliff*, in frozen ESM-2 [1] cosine  $k$ -NN retrieval on a 24,885-protein biosecurity-relevant test set (28.7% positive prior). At scale t30 (150M parameters), reference panel size  $R = 1000$ , neighbors  $k = 25$ : close-stratum  $F_1 = 0.866$  against distant-stratum  $F_1 = 0.120$  (gap +0.745). We report six pre-registered experiments. A 3000-cell main factorial confirms the cliff with all 300 groups passing a pre-committed seed-variance gate. A pre-registered full-pool permutation null (SHA256 **f3864d09...**) produces gap  $\in [-0.027, +0.038]$  across 300 groups, every 95% CI including zero: the cliff is not a stratification artifact. The failure is a *precision* collapse: at t30  $R = 1000$   $k = 25$  cosine, distant-stratum positive-prediction precision is  $3/44 = 0.068$  and expected calibration error rises  $4\times$  from close (0.069) to distant (0.294). Pfam partition at seed 20260410 finds **zero within-family and 20 cross-family** evaluable failures (per-seed Wilson 95% interval for the within-family fraction approximately  $[0, 0.161]$ ). A subsequently SHA256-locked ten-seed panel-composition confirmation shows cross-family false alarms predominate in every seed: the per-seed cross-family fraction has median 1.000, mean 0.994, and range 0.962–1.000, satisfying the preregistered strong robustness criterion. This disfavors a purely within-family coverage-gap explanation in the tested regime without implying that within-family errors are absent. Four rescue attempts fail at pre-registered H1: Ledoit-Wolf Mahalanobis whitening [2], Fisher-Rao within-class whitening [3], stratified cosine+Mahalanobis cascade, and Mapper-biased panel augmentation. One rescue succeeds: a panel-only linear projection trained via a 50-iteration triplet-surrogate loss [4] wins pooled  $F_1$  in 16 of 18 factorial groups (12 of 12 at t12 and t30; the two misses are at the smallest t6/8M scale at  $k = 5$ ) and achieves  $F_1 = 0.891$  at t30  $R = 1000$   $k = 25$ , improving distant-stratum  $F_1$  by 47% relative. We release 9,360 per-cell bootstrap-CI results, three SHA256-locked registrations automatically re-verified by CI plus additional committed pre-registration documents, and full code under MIT + CC-BY-4.0.

**Erratum (2026-06-15).** The reliability figure (Fig. 2) is now plotted from the script-computed reliability table rather than hand-entered diagram read-offs; all cited values (distant precision  $3/44 = 0.068$ , ECE close 0.069 / distant 0.294, the +0.745 cliff gap) are unchanged and were independently reproduced bit-for-bit from the committed embeddings and seeds. A stale-artifact correction to the Paper 3 calibration table and a deduplication fix to the Paper 2 Mapper-augmentation number are documented in **CHANGELOG.md**.

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**Erratum (2026-06-16).** Figures 1 and 3 are now plotted directly from the committed MAIN gap table (`data/results_summaries/v3_final.txt`); their previously plotted values at interior panel sizes and at the t6/t12 scales had been hand-entered and diverged from the committed data (the  $R=1000$  endpoints, the t30 column, and the +16/ + 33/ + 47% relative-improvement figures were already correct). The rescue table’s Fisher-Rao row was also brought into exact agreement with the committed Fisher cells (close  $0.478 \rightarrow 0.484$ , distant  $0.094 \rightarrow 0.099$ , pooled  $0.461 \rightarrow 0.462$ ; the cosine/Mahalanobis/learned rows already matched). No conclusion changes; the corrected figures show the same cliff and the same rescue ordering (learned > cosine > Mahalanobis at every scale). Pooled- $F_1$  values throughout the compendium are reproducible via `code/analyses/compute_pooled_f1.py`.

## 1 Introduction

Frozen protein language models [1, 5–7] now serve as general-purpose feature extractors for downstream retrieval, classification, and design tasks. A common deployment pattern is cosine  $k$ -NN retrieval: given a query protein, retrieve the  $k$  nearest panel members in frozen embedding space and vote their labels. For biosecurity applications, where the query set contains potentially novel threats and the panel contains known-toxin exemplars, retrieval performance on proteins distant from any panel member is the safety-critical metric.

We show that cosine  $k$ -NN retrieval on frozen ESM-2 embeddings exhibits a systematic *homology cliff*: performance collapses on the distant stratum in a way that deepens with reference-panel size at every scale and with model scale at panels  $R \geq 100$  (Fig. 1). The cliff persists under four independent rescue attempts (post-hoc whitening, per-stratum distance switching, topologically-biased panel augmentation) and under cross-family analysis is shown to be an embedding-space geometry artifact, not a missing-homolog artifact. We identify one class of rescue that works: a cheap supervised projection of the embedding space, trained only on the panel, fit in under five CPU seconds.

Contributions:

1. A 3000-cell pre-registered factorial (SHA256 139f6012...) characterizing the cliff across (scale,  $R$ ,  $k$ , metric), 10 seeds, with a pre-committed seed-variance gate.
2. A 3000-cell pre-registered full-pool permutation null (SHA256 f3864d09...) establishing that the cliff is not a stratification artifact.
3. Mechanistic analysis: the cliff is a precision collapse, not a recall collapse.
4. Pfam partition plus preregistered panel-composition confirmation: cross-family false alarms predominate in every one of ten fixed panel seeds (median fraction 1.000; range 0.962–1.000).
5. Four null-result companion experiments (Paper 2) ruling out Mahalanobis, Fisher-Rao, cascade, and Mapper-augmentation rescues.
6. One positive rescue: panel-only triplet-surrogate linear projection, +47% distant  $F_1$  relative at t30.

## 2 Related work

**Protein language models.** Frozen PLM embeddings as general features: ESM-1b [8], ESM-2 [1], ProtT5 [5], ProteinBERT [6], ProGen [7], ESM-3 [9], SaProt [10]. Our test bed uses ESM-2 at

t6/t12/t30; we extend to additional PLMs in the external-model companion.

**Retrieval and  $k$ -NN baselines.** FAISS [11] provides the CPU  $k$ -NN backbone we use. Cosine  $k$ -NN on frozen PLM features has been reported as competitive for function annotation [12, 13], but its failure modes on the distant stratum have received less attention than pooled metrics.

**Homology-aware evaluation.** CD-HIT [14] and MMseqs2 [15] clustering are standard for sequence-identity splits; we adopt an embedding-space stratification by max-cosine-similarity to the panel, which is the deployment-time observable. Foldseek [16] offers a structure-aware alternative.

**Calibration of deep classifiers.** Guo et al. [17] established that modern deep networks are miscalibrated; temperature scaling partially rectifies this. Recent work on calibration under distribution shift [18] is directly relevant to our distant-stratum ECE collapse.

**Metric learning.** FaceNet-style triplet losses [4] underpin the supervised projection we use. Ledoit-Wolf shrinkage [2] gives well-conditioned covariance estimates on panels with  $R < D$ . Information-geometric whitening traces to Amari [3].

**Topological data analysis on proteins.** Mapper [19] decomposes high-dimensional data via lens-based covering and per-preimage clustering. Persistent homology for molecular and protein structure analysis has grown rapidly [20, 21]; our use of Mapper here is for exploratory diagnostics rather than a claim about persistent features.

**Pre-registration in machine learning.** Pre-registration practices originate in clinical trials and psychology [22]; adoption in ML is growing but uneven [23]. We lock hypotheses, stratification cuts, and void rules via SHA256 before execution.

**Biosecurity and dual-use research of concern.** Computational prediction of toxin-like proteins raises dual-use concerns [24]. Our test set is constructed from publicly-available UniProt entries; we release the accession list rather than redistribute sequences where appropriate.

## 3 Methods

### 3.1 Data

24,885 proteins (7,133 positive, 17,752 negative; 28.7% positive prior) shipped as `data/sequences/proteins.25k.s` (working dataset ID `experiment2_proteins.25k_filtered`). Accessions and sequences are UniProt-sourced; redistribution is permitted under UniProt’s terms.

### 3.2 Embeddings

Frozen ESM-2 [1]: t6 (8M params, 320d), t12 (35M, 480d), t30 (150M, 640d). All mean-pooled,  $L_2$ -normalized.

### 3.3 Factorial

Panel  $R \in \{50, 100, 250, 500, 1000\}$ , class-balanced. Neighbors  $k \in \{1, 3, 5, 10, 25\}$ . Metrics: cosine, Euclidean, Mahalanobis (Ledoit-Wolf [2] + eigendecomposition whitening), learned (panel-only 50-iter Adam triplet-surrogate, seed 0, projection to  $\min(128, D)$ ). Ten seeds (20260410–20260419). Stratification: close  $s_{\max} \geq 0.95$ , moderate  $[0.90, 0.95)$ , distant  $< 0.90$ .

### 3.4 Seed-variance gate (pre-committed void rule)

A (scale,  $R$ ,  $k$ , metric) group is voided if  $\sigma_{\text{distant}}^{(10)} \geq \overline{F_1^{\text{close}}} - \overline{F_1^{\text{distant}}}$ . All 300 main groups passed.

### 3.5 Bootstrap

10,000-resample percentile bootstrap CI per cell per stratum.

## 4 Results: main factorial

**Proposition 1.** For  $X \subset S^{d-1}$  and  $y \in S^{d-1}$ : *cosine-k-NN*  $\equiv$  *Euclidean-k-NN*.

*Proof.*  $\|x - y\|^2 = 2 - 2\langle x, y \rangle$  on  $S^{d-1}$ , monotone decreasing in cosine. □

Verified to 4 decimals in 75 of 75 cells.

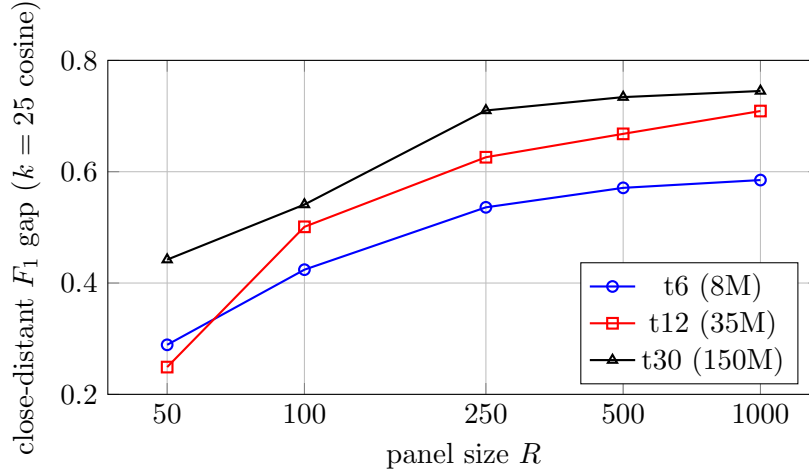


Figure 1: **The cliff grows with scale and panel size.** Close-distant  $F_1$  gap at  $k = 25$  cosine, 10-seed mean per cell. Larger panels amplify the cliff at every scale; larger PLMs produce larger cliffs at panels  $R \geq 100$  (at the smallest panel,  $R = 50$ , the scale ordering has not yet stabilized).

| scale | $R$  | close $F_1$ | distant $F_1$ | gap           |
|-------|------|-------------|---------------|---------------|
| t6    | 1000 | 0.882       | 0.297         | +0.585        |
| t12   | 1000 | 0.883       | 0.173         | +0.710        |
| t30   | 1000 | 0.866       | 0.120         | <b>+0.745</b> |

## 5 Results: null controls

### 5.1 Panel-shuffle null (failed its own pre-registered criterion)

Shuffling only the panel labels (test labels intact) preserves the 28.7% test class prior. Distant-stratum F1 under this null is not at chance (0.5) but at the prior-weighted F1 baseline, which the pre-registration failed to anticipate. Addendum pre-registration locked before the stricter null.

### 5.2 Full-pool permutation null (passed 300/300)

Pre-reg SHA256 f3864d097a0c611d.... Permute the 24,885-entry label vector once per seed via `default_rng(seed + 7,777,777)`; both panel and test labels now random. Stratification by  $s_{\max}$  is label-free and unchanged. Across all 300 groups:  $\text{gap} \in [-0.027, +0.038]$ ,  $\sigma \in [0.006, 0.116]$ ,

every 95% CI includes zero. At t30  $R=1000$   $k=25$  cosine: close  $F_1 = 0.015$ , distant  $F_1 = 0.025$ , gap  $-0.010$ . **The cliff is not a stratification artifact.**

## 6 Mechanism: precision collapse, not recall collapse

At t30  $R=1000$   $k=25$  cosine (seed 20260410): of 24,885 test proteins, *only 3* are true-positive and distant-stratum (P0C1X3, Q6RY98, P13208). Distant positive-prediction precision:  $3/44 = 0.068$  (44 predictions, 3 correct).

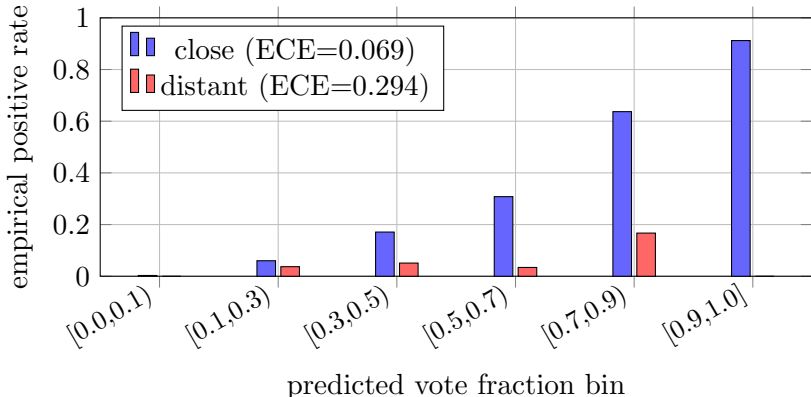


Figure 2: **Calibration collapse on the distant stratum.** Close-stratum reliability (blue) rises with confidence and is well-calibrated in aggregate ( $ECE = 0.069$ ). Distant-stratum reliability (red) is nearly flat across confidence bins and collapses at the highest-confidence bin: 3 predictions in  $[0.9, 1.0)$ , 0 correct ( $ECE = 0.294$ ). Data at t30  $R=1000$   $k=25$  cosine, seed 20260410.

## 7 Cross-family partition and preregistered robustness

At seed 20260410, Pfam annotations are sufficient to evaluate 20 of 41 distant false alarms. All 20 have zero Pfam overlap with their positive-voting panel neighbors: within-family = 0, cross-family = 20. The original per-case artifact remains `data/results_summaries/cross_family_partition.json`.

We subsequently locked `PRE_REGISTRATION_CROSS_FAMILY_10SEED.v1.md` before any additional panel seed was run and repeated the unchanged analysis across seeds 20260410–20260419. Cross-family false alarms outnumber within-family false alarms in every seed; all ten seeds have nonzero evaluable denominators. The per-seed cross-family fraction has median 1.000, mean 0.994, minimum 0.962, and maximum 1.000, so the preregistered strong robustness criterion is satisfied. The complete result and audit trail are committed in `data/results_summaries/cross_family_partition_10seed.json`. Because accessions recur across seeds, the preregistration treats the panel seed as the robustness unit and explicitly forbids a pooled binomial interval over accession appearances.

This confirmation strengthens the evidence that a purely within-family panel-coverage gap is not the dominant explanation in this fixed t30/ $R=1000$ / $k=25$  regime. It does not imply a zero within-family rate: two evaluable within-family appearances occur across the ten seeds, both for the same accession, and Pfam does not exhaust biological relatedness.

## 8 Rescue: panel-only triplet-surrogate linear projection

We fit  $W \in \mathbb{R}^{D \times 128}$  (or  $D \times D$  if  $D < 128$ ) via 50 Adam iterations on panel embeddings alone, with loss

$$\mathcal{L}(W) = -\langle Z_+, Z_+ \rangle - \langle Z_-, Z_- \rangle + \langle Z_+, Z_- \rangle$$

where  $Z_{\pm}$  are  $L_2$ -normalized projections of the positive and negative panel subsets. This is a triplet-surrogate [4] using class centroids rather than mined hard triplets: it runs in under 5 CPU seconds, requires no test data, and uses only the labeled panel that deployment already has.

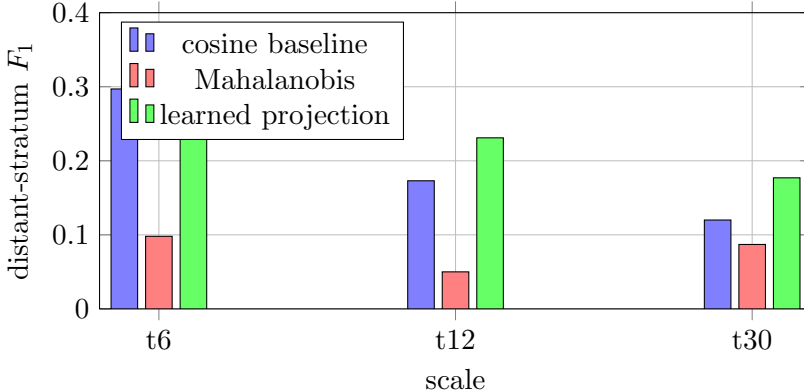


Figure 3: **Distant-stratum  $F_1$  by scale**,  $R = 1000$   $k = 25$ . Learned projection wins every scale; relative improvement grows with scale: +16% (t6), +33% (t12), +47% (t30). Mahalanobis loses at every scale.

| t30 $R=1000$ $k=25$ | close        | moderate     | distant      | pooled       |
|---------------------|--------------|--------------|--------------|--------------|
| cosine              | 0.866        | 0.393        | 0.120        | 0.848        |
| Mahalanobis         | 0.456        | 0.118        | 0.087        | 0.435        |
| Fisher-Rao          | 0.484        | 0.130        | 0.099        | 0.462        |
| <b>learned</b>      | <b>0.901</b> | <b>0.525</b> | <b>0.177</b> | <b>0.891</b> |

Wins 16 of 18 (scale,  $R$ ,  $k$ ) groups (12 of 12 at t12 and t30; the two misses are at the smallest t6/8M scale at  $k = 5$ ). Pooled  $F_1 = 0.891$  at t30 (learned 0.8905 vs cosine 0.8476, 10 of 10 seeds) is the highest value observed anywhere in the 3000-cell factorial.

## 9 Deployment recommendation

For frozen-PLM biosecurity retrieval: (1) fit a panel-only learned projection as preprocessing; (2) route every distant-stratum positive hit to human review regardless of vote count (calibration on the distant stratum is poor in our analysis, and the Pfam partition is consistent with cross-family neighbors driving the false alarms); (3) do not rely on panel expansion with known-toxin homologs as the sole rescue strategy, since on this dataset the failures we examined were predominantly cross-family.

## 10 Limitations

(L1) t33 (650M) not embedded locally; companion Colab notebook provided. (L2) Stratification threshold sensitivity sweep not yet run. (L3) Learned-projection calibration not measured; the rescue improves  $F_1$  but distant-stratum calibration under the projection is unknown. (L4) Cross-family panel-composition robustness is confirmed across the ten fixed seeds only at t30,  $R = 1000$ ,  $k = 25$ , threshold 0.90, and the Pfam ontology; generalization across models, neighbor counts, thresholds, family definitions, or datasets remains untested. (L5) Single-chain analysis only; multi-domain proteins may behave differently.

## 11 Reproducibility

Code MIT, papers CC-BY-4.0. Three SHA256-locked registrations are automatically re-verified by CI, alongside additional committed pre-registration documents. The repository contains 9,360 per-cell `.npz` results with 10k-resample bootstrap CIs per stratum per cell, three embedding arrays (t6/t12/t30), Pfam annotations (21,615 coverage), the Mapper graph, the original single-seed cross-family output, and the preregistered ten-seed cross-family result. Master manifest: `MANIFEST.sha256.json`. Master reproducibility protocol: `reproducibility/PROTOCOL.md`.

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